

Chao Huang*, Anas Ali Alhur, Mehreen Azam and Salman Bin Naeem

The Challenge of Academic Integrity in the Age of Generative Artificial Intelligence

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Abstract: This study aims to understand the factors influencing academic integrity in the age of generative artificial intelligence (GenAI) through the lens of the Theory of Planned Behavior (TPB). A seven-factor measurement model was developed, hypothesizing the relationship between TPB constructs – e.g., perceived behavioral control (PBC), past behavior (PB), moral obligations (MO), social norms (SN), information literacy (IL)—and GenAI adoption. It was tested using structural equation modeling (SEM). The findings showed that both PBC ($\beta = 0.223$, CR = 4.234, $p < 0.05$) and SN ($\beta = 0.508$, CR = 5.644, $p < 0.05$) significantly increased behavioral intention. MO ($\beta = -0.273$, CR = -4.234, $p < 0.05$) and IL ($\beta = -0.253$, CR = -3.386, $p < 0.05$) helped uphold academic integrity. A strong association was found between behavioral intention (BI) and GenAI adoption (AD) for academic dishonesty ($\beta = 0.651$, CR = 11.780, $p = 0.000$). The findings suggest that protecting academic integrity needs a multi-tiered approach, including implementing AI literacy programs, research, and critical thinking training, implementing strong plagiarism detection tools, developing and maintaining comprehensive and adaptable guidelines for the ethical use of GenAI, citation management training, and cultivating a culture valuing academic honesty and originality.

Keywords: academic misconduct; generative artificial intelligence (GenAI); ChatGPT; AI literacy; academic integrity; ethical implications

1 Introduction

Generative artificial intelligence (GenAI), like AI chatbots, uses large volumes of text data to generate human-like content. This is possible due to its generative pre-trained transformer (GPT) architecture (Akbar et al 2023). These models are trained on massive datasets. They are optimized for specific natural language processing (NLP) tasks like language synthesis and summarization. ChatGPT, a popular GenAI tool, has grown rapidly, reaching over 100 million users, making it one of the fastest-growing applications (Van Bulck and Moons 2024). The rise of ChatGPT and similar LLMs has sparked interest in their capabilities and impacts. Researchers and developers are exploring their potential benefits and risks.

Generative AI (GenAI) is transforming every sector, including higher education and healthcare. AI is already improving healthcare delivery, paving the way for medical education (Kamal 2024). Numerous discussions and predictions of AI in medical education have focused on its potential capabilities, applications, and concerns (Civaner et al. 2022; Watson 2024). In nursing education, AI can enhance training with personalized, data-driven learning. AI prepares nurses for an evolving, tech-driven healthcare system. Taira et al (2023) investigated ChatGPT's performance in passing the National Nursing Exam in Japan from 2019 to 2023. The findings indicated that ChatGPT scored 80 % in 2019 and passed all subsequent years' exams. However, a high success rate of ChatGPT on standardized exams does not show that AI use can supplement nursing education by providing access to required theoretical and practical information. It is essential to consider the limitations of AI in replicating the critical thinking and clinical decision-making skills essential for nurses (Taira et al 2023).

The objective of the study is to determine the factors that influence behavioral intentions to adopt generative artificial intelligence (GenAI) tools for academic dishonesty.

2 Literature Review

The use of generative AI tools in nursing education has multiple challenges such as concerns over academic integrity, the possibility of plagiarism and copyright violations, and the potential for disseminating inaccurate information.

***Corresponding author: Chao Huang**, School of Art, Tianjin University of Technology, Tianjin, 300384, China, E-mail: chao_huang2025@126.com

Anas Ali Alhur, Department of Health Information Management and Technology, College of Public Health, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia, E-mail: anas.ali.alhur@gmail.com

Mehreen Azam, Higher Education Department, Government of the Punjab, Bahawalpur, Pakistan, E-mail: mehreenazam100@gmail.com

Salman Bin Naeem, Department of Information Management, The Islamia University of Bahawalpur, Bahawalpur 63100, Pakistan, E-mail: salman.naeem@iub.edu.pk

The excessive reliance on AI generative tools may significantly impede students' problem-solving and critical thinking skills (Chen and Chen 2023). However, AI generative tools can supplement but not replace these critical skills. While ChatGPT is extensively trained on a massive amounts of data, it is important to acknowledge the potential for errors in its training process (Lee 2023). Because the training data itself may contain incorrect information, there is a risk that the system may be unable to deliver the latest and most accurate information, thus compromising its reliability (Gupta et al. 2023). Therefore, relying solely on generative AI information for clinical decisions is potentially dangerous at this point when AI system is evolving and learning.

Moreover, students' use of generative AI tools for assignment writing raises concerns about academic integrity. There are multiple factors that contribute to students' plagiarism practices. For example, intentional plagiarism occurs when students experience time constraints and perceive pressure to complete an academic task (Fischer and Zigmond 2011; Azam and Naeem 2022). Some other risks and limitations associated with the use of AI generative tools are ethical issues, dubious accuracy, and citation issues (De Gagne et al. 2023). In conclusion, integrating GenAI into education has the potential to significantly improve learning outcomes (Irwin et al 2023). To maximize the benefits of GenAI while maintaining educational standards, scope, and relevance, it is essential to integrate these tools after careful consideration, taking into account both the opportunities and challenges presented by GenAI (Sallam 2023). It demands developing well-defined institutional policies, the provision of clear instructions to students on the responsible use of GenAI, and the implementation of detective AI tools (Borsari et al. 2018; Ko and Song 2024). Thus, in order for successful and ethical integration, it is important to understand students' perspectives on GenAI tools adoption, as well as factors that influence their misuse of GenAI in completing educational tasks.

Researchers have developed numerous theoretical frameworks and models to understand the adoption and usage behavior of information technology and systems. These models and theories include "the diffusion of innovation theory" that explains how new ideas, technologies, or products spread through a population over time (Rogers 1995). The motivational model primarily focuses on human motivation in general and understanding these motivations can help explain why some individuals are more eager to adopt new technologies than others (Maslow 1943). The social cognitive theory suggests that individuals learn to use and adopt new technologies by observing others, interacting with social networks, and through social reinforcement (Bandura 1991). The model of personal computer utilization

focuses on factors that influence individuals' use of personal computers (Thompson et al 1991). Innovation diffusion theory focuses on the spread of innovations within social systems (Ryan and Gross 1943). The theory of reasoned action suggests that intentions to use technology are influenced by attitudes towards technology and social norms regarding its use (Ajzen and Fishbein 1977). Technology acceptance model (TAM) emphasizes key factors, such as perceived usefulness and perceived ease of use, that influence technology acceptance (Devis 1989). The unified theory of acceptance and use of technology (UTAUT) suggests that factors including performance expectancy, effort expectancy, social influence etc., influence technology acceptance and use (Venkatesh et al. 2003). It was developed by Venkatesh et al. (2003), and its extension, UTAUT2 (Venkatesh et al 2012), which includes constructs such as hedonic motivation, price value, and habit, provides a comprehensive framework for understanding new technology adoption behavior.

In 1977, Ajzen and Fishbein introduced the theory of reasoned action, which claims that an individual's intentions to engage in a particular behavior is determined by their attitude towards the behavior and subjective norms. This theory was expanded by incorporating perceived behavioral control, leading to the development of the theory of planned behavior (TPB). According to the TPB, understanding behavior requires examining the intentions, which are influenced by three key factors: attitudes towards the behavior, subjective norms, and perceived behavioral control (Ajzen 1985). Previously, several studies have successfully applied the TPB to understand technological adoption behavior (Yi et al. 2006; Cronan and Al-Rafee 2008; Alwahaishi and Snášel 2013; Hsiao and Tang 2014). Although TPB has been applied to study various behavioral domains, its specific application to understanding the factors influencing academic integrity within the context of GenAI tools adoption represents a novel contribution to the literature. Therefore, recognizing the applicability of this framework in the context of this study, we adopted the extended theoretical framework incorporating key constructs from the TPB – perceived behavioral control (PBC), moral obligations (MO), past behavior (PB), and subjective norms (SN) – along with an additional construct: information literacy (IL) (as shown in Figure 1).

The relationships between the TPB constructs (e.g., past behavior, moral obligations) and behavioral intention to adopt AI were hypothesized based on the relevant literature (Yang et al. 2021). It also supported the research model's predicted positive/negative relationships (Figure 1).

Perceived Behavioral Control (PBC) is how easy or hard someone thinks a behavior is. People with high PBC believe they can do the behavior well. They also expect fewer obstacles in their way. In contrast, low PBC means a person

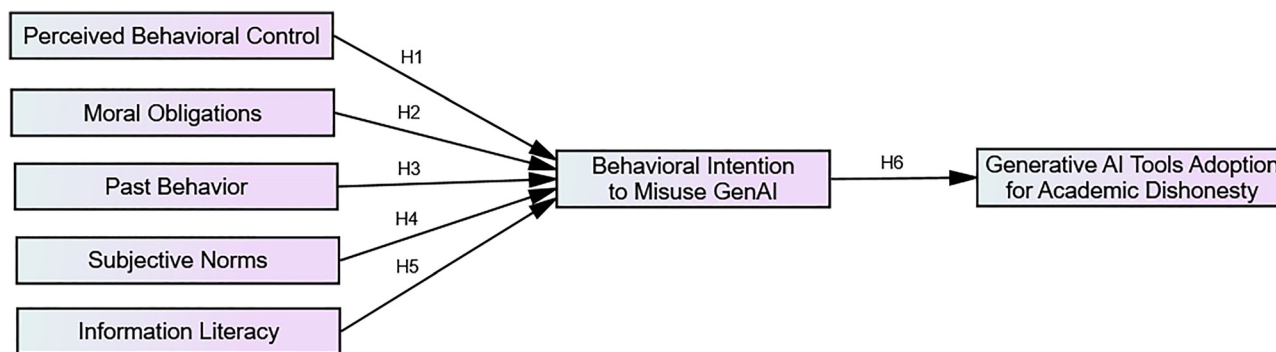


Figure 1: Proposed research model hypothesizing the relationship between TPB constructs and behavioral intention of AI adoption.

doubts their ability to perform the action (Ajzen and Fishbein 1977; Conner and Armitage 1998; Yi et al. 2006). The literature indicate that individuals with higher PBC are more likely to form strong intentions to carry a behavior (Ajzen 1991; Morris and Venkatesh 2000; de Andrés-Sánchez and Gené-Albesa 2023). Students with higher PBC may perceive it easier to use GenAI to complete academic assignments without proper acknowledgement. This PBC could lead to a stronger intention to misuse GenAI tools for academic purposes. Therefore, we propose the following hypothesis:

H1: Perceived behavioral control significantly and positively influences the behavioral intention to adopt GenAI tools for academic dishonesty.

Moral obligation (MO) refers to an individual's subjective sense of responsibility to either participate in or avoid a specific activity (Ajzen 1991). Moral reasoning is the cognitive process through which individuals differentiate between morally right and wrong actions in a particular situation (Rest et al 1997). The revised version of the theory of planned behavior incorporates "moral obligation" as an important construct. This modification enhances the theory's effectiveness in predicting dishonesty in academia (Rujoiu and Rujoiu 2014).

Thus, individuals with a strong sense of moral obligation are likely to perceive the misuse of GenAI tools as ethically wrong (Yang et al. 2021), thereby reducing their intention to engage in such behavior (Newell and Dahlenburg 2024). This aligns with the theory of planned behavior (TPB), which suggests that moral obligation can act as a deterrent to unethical actions (Ajzen 1991; Rujoiu and Rujoiu 2014). Therefore, we propose the following hypothesis:

H2: Moral obligations significantly and negatively influence the behavioral intention to adopt GenAI tools for academic dishonesty.

Past behavior (PB) pertains to an individual's prior activities or conduct in a comparable circumstance. It is regarded as a

significant predictor of future actions (Ajzen 1991). Past actions can shape an individual's intentions and probability of repeating a behavior. Examining an individual's past performance provides insight into their potential future behavior. Few studies have reported past behavior as a significant predictor of indulging in academic dishonesty (Harding et al. 2007; Imran and Nordin 2013). It suggests that individuals who have engaged in academic dishonesty in the past are more likely to do so again.

Thus, individuals who have previously misused technology or engaged in academic dishonesty are more likely to form intentions to repeat such behavior, as past actions create a behavioral precedent. This is consistent with the Theory of Planned Behavior (TPB), which acknowledges the role of habit and past behavior in shaping intentions (Ajzen 1991). Therefore, we propose the following hypothesis:

H3: Past behavior significantly and positively influences the behavioral intention to misuse GenAI tools for academic dishonesty.

Subjective norms (SN) refer to the societal standards of individuals regarding a particular action, as influenced by others. Students who think that social norms encourage cheating are more likely to engage in cheating compared to other students (Ajzen 1985). This notion, also sometimes referred to as injunctive norms, reflects the perception of other people's approval or disapproval. The literature indicated that peer-related factors have a significant impact on behavioral intention (de Andrés-Sánchez and Gené-Albesa 2023). McCabe, Trevino and Butterfield's (2002) study of nine colleges found that peer-related contextual factors have a significant influence on students' behavior. Thus, if students perceive that their peers or social groups engage in misusing GenAI tools, they are more likely to do so. This notion aligns with the Theory of Planned Behavior (TPB), which posits that subjective norms are a key determinant of behavioral intentions. Therefore, we propose the following hypothesis:

H4: Subjective norms significantly and positively influence the behavioral intention to adopt GenAI tools for academic dishonesty.

According to the Association of College and Research Libraries (ACRL) (2016), information literacy (IL) includes the cognitive skills necessary to recognize, locate, evaluate, and effectively utilize information. Students with low information literacy tend to passively consume information from the Internet. A deficiency in critical thinking skills can also be linked to poor IL. It can result in poor information seeking behavior and can also increase the likelihood of engaging in academic dishonesty through the misuse of technology such as GenAI (Khan et al 2021; Islam et al 2022). The literature suggests that IL is a strong predictor of behavioral intention (Lund et al 2023). Individuals with IL skills are more able to adopt, use, and utilize GenAI tools responsibly and effectively than those with poor information literacy skills (Jones-Jang et al 2019; Pennycook et al. 2021). They are also better equipped to evaluate the ethical implications of using GenAI tools and are less likely to misuse them (Lund et al 2023). Therefore, we propose the following hypothesis:

H5: Information literacy significantly and negatively influences the behavioral intentions to adopt GenAI tools for academic dishonesty.

Behavioral intention (BI) is a key concept in theories of human behavior, such as the theory of planned behavior (TPB). It refers to how likely a person is to perform a specific action (Ajzen 1991; Venkatesh et al. 2003; Venkatesh 2022). According to TPB, behavioral intention strongly influences the adoption and use of GenAI (Andrews et al 2021; Bilquise et al 2023). Thus, if students have a firm behavioral intention to misuse GenAI tools, they are likelier to adopt them for academic dishonesty. Therefore, we propose the following hypothesis, which is aligned with the constructs of the TPB and provides empirical support for the relationship between behavioral intention and adoption (Ajzen 1991; Venkatesh et al. 2003; Venkatesh 2022).

H6: Behavioral intention to misuse generative AI (GenAI) tools has a significant and positive influence on the adoption of GenAI for academic dishonesty.

3 Methodology

3.1 Participants and Procedures

A cross-sectional survey design was employed to collect data from the participants. The study participants included enrolled students in the Bachelor of Science in Nursing (BSN)

program at the four participating institutions in the South Punjab region of Pakistan. The study included: College of Nursing, Islamia University of Bahawalpur; KIMS Mustafa Kamal Institute of Nursing and Medical Sciences, Vehari; New Life Institute of Nursing, Multan; and College of Nursing, Bahawalpur.

A cross-sectional survey was used. It collected data on factors influencing GenAI use for academic dishonesty. The collected data is used to test the relationship based on the theory of planned behavior (TPB). The nursing population was selected for several reasons, including that nursing education increasingly uses AI for learning and skill development. Nursing requires high ethical standards due to its impact on patient care. Thus, insights from nursing students can provide broader implications for academic integrity across various disciplines, including social sciences, sciences, and the arts.

The South Punjab region has a diverse socio-cultural background and has well-established nursing institutions, which is why it was selected.

3.2 Research Tool

The questionnaire was developed based on the Theory of Planned Behavior (TPB) and generative AI adoption literature (Ostick et al. 2025; Park et al 2024; Topaz et al. 2024; Yang et al. 2021). It had three sections. The first section collected data on demographic information, which included gender, age, institution type (public/private), and program year. The second section of the questionnaire was based on statements on seven constructs adopted from the TPB (Figure 1). For example, the construct “perceived behavioral control” was measured using five statements, and “moral obligations” also comprised of five statements. The sub-scale on the construct “past behavior” comprised of six statements, “subjective norms” contained three statements, the construct “information literacy” consisted of five measurable statements, and “behavioral intention” included four measurable statements. The third part of the questionnaire included questions measuring participants’ perceptions and behavioral intentions towards “generative AI tools adoption.”

3.3 Validity and Reliability Measures

The questionnaire was pre-tested by two academicians in the field of nursing, including a nursing practitioner. The proposed changes were incorporated into the scale such as removing double barreled questions, avoiding technical

jargon, and shuffling a few questions from one sub-scale to another. Following these changes, the questionnaire was pilot tested on a cohort of thirty nursing students to assess its reliability and clarity in a real-setting. Feedback from the pilot phase confirmed the questionnaire's appropriateness for the target population after a few minor adjustments to working and formatting based on participant suggestions.

Cronbach's alpha was applied to measure the questionnaire's internal consistency of each subscale and reliability, which provided a score of 0.898 for the thirty-two items across seven constructs, indicating the questionnaire's good reliability. The Cronbach's alpha coefficients values for each construct estimated in this study are reported in Table 1.

The Cronbach's alpha values for three measures fall within the 0.6 range, which is considered acceptable for exploratory research (Streiner et al 2024; Streiner 2003).

Another area of focus was univariate and multivariate normality assessment. Assessing normality is crucial in SEM analysis. It impacts parameter estimates, standard errors, and hypothesis testing. We checked skewness and kurtosis using maximum likelihood estimation (MLE) to assess normality. All variables met skewness criteria (<3) (Kline 2023), and kurtosis values were also acceptable (<7). Multivariate kurtosis was within limits (<5) (Yuan et al 2005). The findings suggest reasonable symmetry. Both univariate and multivariate normality assumptions were met (West et al 1995).

3.4 Sample Size Calculation and Method

We used purposive sampling to select participants from the participating colleges. During participant recruitment, respondents were explicitly asked: "Have you ever used AI

tools (e.g., ChatGPT, Gemini, DeepSeek) for academic assignments?" Only those participants who answered "Yes" were included in the final sample, which ensured the recruitment of participants who had direct experience with AI-assisted academic work. Moreover, to select the minimum required sample size, we used *a-Priori* Sample Size Calculator (Soper 2020). An average effect size of 0.3 (small: 0.1, medium: 0.3, large: 0.5) was set while calculating the sample size. Moreover, considering the model's complication (seven constructs and 32 observed variables), a preferred statistical power of 0.95 (conventionally, >0.80) was used (Cohen 1988), as well as an alpha coefficient value set at 0.05. We calculated a minimum sample size of 212 participants. To ensure sufficient statistical power and minimize the margin of error, we continued data collection beyond the minimum sample size of 212 participants.

3.5 Data Collection Process

The questionnaire was administered in English, which is the de facto official language of Pakistan and the medium of instruction in the participating institutions. There were more than 1,500 nursing students enrolled in the participating institutions. In total, 600 copies of the questionnaire along with the consent form to participate in the study were administered among participants between June 2024 and August 2024. A combination of online and paper-based questionnaires was used to collect the data. A five-point Likert type scale (from Strongly Disagree to Strongly Agree) was used to obtain the responses. The researchers personally visited the participating colleges to collect the data. The heads of participating colleges served as study facilitators, assisting in the process of data collection at their respective institutions. The participation of the respondents and study facilitators was voluntary.

Regarding the response rate, out of the 600 questionnaires distributed, 331 were returned (55.16 % return rate). Of these, 314 responses were valid for data analysis, resulting in a response rate of 52.33 %. Seventeen questionnaires were excluded due to missing data or multiple responses to single-answer questions. The response rate was boosted by three follow-up requests, each fifteen days apart.

3.6 Data Analysis Procedure

The data was analyzed using "IBM's Statistical Package for Social Sciences" (SPSS v26 by IBM) and Analysis of Moment Structure (AMOS). An alpha level of 0.05 was used to

Table 1: Cronbach alpha coefficients value.

Sr. no	Constructs	No. of items	Cronbach's alpha
1	Perceived behavioral control (PBC)	5	0.852
2	Moral obligations (MO)	5	0.664
3	Past behavior (PB)	6	0.740
4	Subjective norms (SN)	3	0.623
5	Information literacy (IL)	5	0.621
6	Behavioral intention (BI)	4	0.721
7	Generative AI tools adoption (AD)	4	0.854

$\alpha \geq 0.9$ = Excellent, $0.7 \geq \alpha < 0.9$ = Good, $0.6 \geq \alpha < 0.7$ Acceptable, $0.5 \geq \alpha < 0.6$ = Poor (Streiner et al 2024; Streiner 2003).

determine statistical significance for hypothesis testing. Demographic information is presented as frequencies, percentages, mean, and standard deviation. The proposed conceptual model, based on the Theory of Planned Behavior (Figure 1), was tested using Structural Equation Modeling (SEM) (Figure 2).

3.7 Ethical Approval

The study strictly adhered to established ethical guidelines for research involving human participants. All procedures, including data collection, participant recruitment, and confidentiality protocols, complied with institutional and international ethical standards.

4 Data Analysis and Interpretation

4.1 Demographic Information

Of the 314 respondents, the majority of respondents (62.1 % or 195) were female, while (37.9 % or 119) were male. The majority of respondents fell within the 18–25 age range. Specifically, 147 respondents (46.8 %) were aged 21–25, and 138 (43.9 %) were under 20. Only twenty-nine respondents (9.2 %) were aged 26–30. The largest group of participants (127, or 40.4 %) were in their first year of nursing education. Another 127 students (40 %) were in their second year. Third-year students comprised fifty-six respondents (17.8 %), and the fourth year had the fewest respondents with twenty-five (8 %). In our cohort, a higher amount (163, or 51.9 %) of the nursing students were enrolled in public sector nursing colleges, with the remainder (151, or 48.1 %) enrolled in private sector nursing colleges.

4.2 Descriptive Statistics of Generative AI Tools Adoption

The respondents were asked five questions in order to record their response on the construct “perceived behavioral control” (as shown in Table 1). Of these five statements, only two items received a mean value around 4, showing that the majority of the respondents agreed that “I have the resources necessary to copy from other sources without giving proper credit” (Mean = 3.68, SD = 1.091) and “copying from other sources and not appropriately citing these sources in an assignment would be difficult” (Mean = 3.64, SD = 0.940).

The respondents were asked five questions relating to “moral obligations.” Of these five statements, only two statements received a mean score of around 4, indicating that the majority of the respondents agreed that “I feel a strong sense of responsibility to uphold honesty in their academic career” (Mean = 3.85, SD = 1.068), and “they think that academic dishonesty is wrong and goes against my ethics” (Mean = 3.70, SD = 1.066).

A set of six statements regarding “past behavior” were asked. Of these six statements, only two statements received a mean value of around 4, showing that the majority of the respondents agreed that “I have never engaged in any form of cheating or plagiarism during their academic career” (Mean = 3.90, SD = 0.971) and “I have occasionally copied someone else’s work without giving proper credit” (Mean = 3.87, SD = 1.027).

Subjective norms were measured using three statements (as shown in Table 1). The majority of the respondents agreed that sometimes “many students in my program use uncredited digital sources, which tempts me to do the same” (Mean = 3.53, SD = 1.183). On the other hand, two items received a mean value of around 3, showing that the majority of the respondents neither agreed nor disagreed with these statements i.e., “I think most people who are important

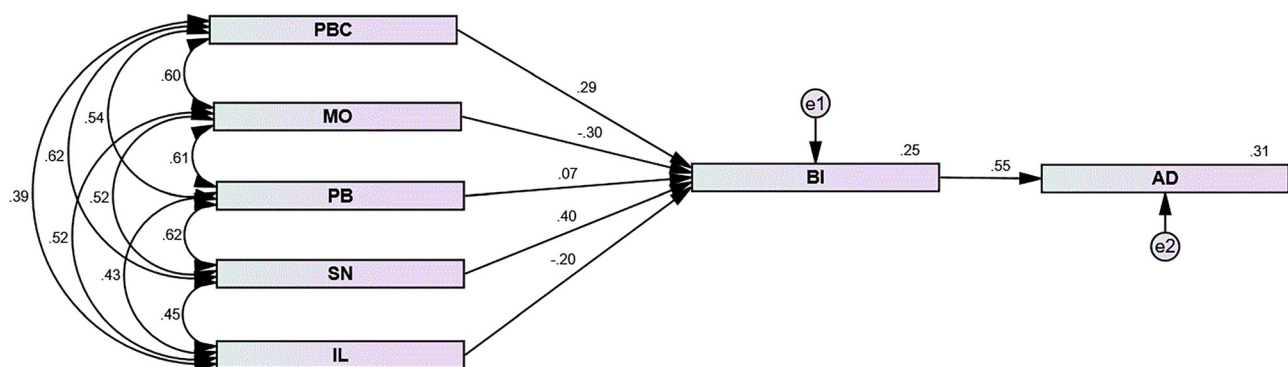


Figure 2: The structural equation model (SEM) estimates students’ generative AI tools adoption. Perceived behavioral control (PBC), moral obligations (MO), past behavior (PB), subjective norms (SN), information literacy (IL), behavioral intention (BI), and generative AI tools adoption (AD).

to me think I should copy others' works from a digital source (e.g., the Internet, ChatGPT) and use them in my own assignments without acknowledgement" (Mean = 3.19, SD = 1.078), etc.

A set of five questions were asked to examine information literacy (IL) skills. All statements had mean scores ranging from 3.69 and 3.90, showing that most respondents agreed with them and possessed good IL skills.

The behavioral intention of the respondents towards adopting generative AI tools was measured using four statements. All questions had mean scores ranging from 3.44 to 3.48, indicating that most respondents neither agreed nor disagreed with the statements.

A set of four questions were asked regarding "generative AI tools adoption." All of these questions received a mean score of around 3, indicating that most respondents neither agreed nor disagreed with the statements (as shown in Table 1).

4.3 Structural Equation Modelling

The seven-factor measurement model was computed using SPSS and was estimated with the help of structural equation modelling (SEM) based on 32 valid items.

4.3.1 Correlation Among Latent Variables

The correlation values indicate that PBC is strongly correlated with MO ($\beta = 0.597$), PB ($\beta = 0.544$), SN ($\beta = 0.618$) and IL ($\beta = 0.395$). The MO is also strongly correlated with PB ($\beta = 0.609$), SN ($\beta = 0.524$), and IL ($\beta = 0.517$). The PB has a positive correlation with SN ($\beta = 0.621$) and IL ($\beta = 0.429$). The SN is also positively correlated with IL ($\beta = 0.447$). Overall, all the constructs (PBC, MO, PB, SN, and IL) are positively correlated in the measurement model with either moderate or strong strength of correlation (Figure 2).

Moreover, the squared multiple correlation (R-squared) indicates that BI accounts for 25 % of the variance in generative AI tool adoption (AD) explained by the latent constructs: Perceived Behavioral Control (PBC), Moral Obligations (MO), Past Behavior (PB), Subjective Norms (SN), and Information Literacy (IL).

4.3.2 Estimation of Covariances

The findings of covariance measurement among exogenous variables show a significant positive correlation between PBC and MO ($\beta = 8.898$, CR = 9.067, $p < 0.05$), PB ($\beta = 9.893$, CR = 8.454, $p < 0.05$), SN ($\beta = 6.477$, CR = 9.296, $p < 0.05$), and IL ($\beta = 4.191$, CR = 6.493, $p < 0.05$). MO has a significant positive correlation with PB ($\beta = 9.448$, CR = 9.204, $p < 0.05$), SN

($\beta = 4.685$, CR = 8.209, $p < 0.05$), and IL ($\beta = 4.684$, CR = 8.125, $p < 0.05$). PB has a statistically positive correlation with SN ($\beta = 6.774$, CR = 9.332, $p < 0.05$) and IL ($\beta = 4.746$, CR = 6.981, $p < 0.05$). SN has a positive correlation with IL ($\beta = 2.852$, CR = 7.226, $p < 0.05$) as shown in Table 2.

4.3.3 Validation of the Hypotheses

The findings of the standardized estimation of regression weight show that PBC ($\beta = 0.223$, CR = 4.234, $p < 0.05$) and SN ($\beta = 0.508$, CR = 5.644, $p < 0.05$) have a significant positive influence on BI. Therefore, hypothesis 1: PBC has a significant and positive influence on BI, and hypothesis 4: SN has a significant and positive influence on BI, with both statistically accepted (as shown in Table 3). On the other hand, MO ($\beta = -0.273$, CR = -4.234, $p < 0.05$) and IL ($\beta = -0.253$, CR = -3.386, $p < 0.05$) exert a significant negative influence on BI. Therefore, the proposed hypothesis 2: MO significantly and negatively influences the BI to misuse GenAI tools for academic dishonesty, and hypothesis 5: IL significantly and negatively influences the behavioral intention, are statistically accepted. Moreover, we found that PB has a non-significant positive influence on BI ($\beta = 0.053$, CR = 1.035, $p = 0.301$). Thus, hypothesis 3 is rejected. On the other hand, we found that BI has a significant positive influence on generative AI tools adoption ($\beta = 0.651$, CR = 11.780, $p = 0.000$). Thus, hypothesis 6 is accepted (as shown in Tables 3 and 4).

4.3.4 Model Fit Indices

To evaluate model fit, several indices were examined, including χ^2 , RMSEA, GFI, NFI, CFI, TLI, and CFI (Kline 2023; Bentler and Bonett 1980). The goodness-of-fit indices suggest an acceptable fit for the model: $\chi^2 = 4.075$, df = 5, $p = 0.001$; GFI = 0.982, AGFI = 0.901, NFI = 0.977, RFI = 0.902, TLI = 0.924, and CFI = 0.982, RMSEA = 0.099. The Chi-Square and RMSEA values are slightly above the recommended threshold, but considering the other fit indices (CFI, TLI, GFI, and NFI), the model can be considered to have an acceptable fit (Table 5).

5 Discussion

The theory of planned behavior (TPB) provides a comprehensive framework to understand students' adoption of novel technologies like generative AI tools in healthcare settings. Previously several studies adopted TPB to explain technological adoption behavior (Ajzen 1985; Morris and Venkatesh 2000; Yi et al. 2006; Harding et al. 2007; Cronan and Al-Rafee 2008; Alwahaishi and Snášel 2013; Wu et al 2017). Recognizing the importance of understanding

Table 2: Descriptive statistics of generative AI tools adoption.

	Statements	Mean	Std. deviation	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
Perceived behavioral control								
PBC1	"If I wanted to, I could easily use generative AI tools (e.g., ChatGPT, Gemini) to copy from other sources without proper attribution."	3.43	1.132	14 (4.5 %)	74 (23.6 %)	36 (11.5 %)	144 (45.9 %)	46 (14.6 %)
PBC2	"I have the skills and knowledge needed to use generative AI tools (e.g., ChatGPT, Gemini) to copy from sources without proper attribution."	3.46	1.078	11 (3.5 %)	72 (22.9 %)	32 (10.2 %)	161 (51.3 %)	38 (12.1 %)
PBC3	"I have the resources necessary to copy from other sources without giving proper credit."	3.68	1.091	11 (3.5 %)	55 (17.5 %)	23 (7.35 %)	161 (51.3 %)	64 (20.4 %)
PBC4	"I am confident that I can use AI chatbots effectively to copy from other sources without citing these sources."	3.39	1.031	7 (2.2 %)	81 (25.8 %)	35 (11.1 %)	164 (52.2 %)	27 (8.6 %)
PBC5	"I think for me to copy from other sources and not appropriately cite these sources in an assignment would be difficult."	3.64	0.940	7 (2.2 %)	47 (15.0 %)	33 (10.5 %)	193 (61.5 %)	34 (10.8 %)
Moral obligations								
MO1	"I feel guilty for copying a significant portion from my previous work without citation."	3.46	1.108	7 (2.2 %)	80 (25.5 %)	39 (12.4 %)	136 (43.3 %)	52 (16.6 %)
MO2	"I think that academic dishonesty is wrong and goes against my ethics."	3.70	1.066	11 (3.5 %)	45 (14.3 %)	37 (11.8 %)	154 (49.0 %)	67 (21.3 %)
MO3	"I think it would be morally wrong to copy information from internet sources."	3.33	1.121	15 (4.8 %)	64 (20.4 %)	90 (28.7 %)	92 (29.3 %)	53 (16.9 %)
MO4	"I think that academic integrity is essential for personal growth and learning."	3.11	1.102	20 (6.4 %)	96 (30.6 %)	44 (14.0 %)	136 (43.3 %)	18 (5.7 %)
MO5	"I feel a strong sense of responsibility to uphold honesty in my academic work."	3.85	1.068	19 (6.1 %)	14 (4.5 %)	49 (15.6 %)	144 (45.9 %)	88 (28.0 %)
Past behavior								
PB1	"I have never engaged in any form of cheating or plagiarism during my academic career."	3.90	0.971	4 (1.3 %)	34 (10.8 %)	38 (12.1 %)	150 (47.8 %)	88 (28.0 %)
PB2	"I have occasionally copied someone else's work without giving proper credit in the past."	3.87	1.027	8 (2.5 %)	37 (11.8 %)	30 (9.6 %)	153 (48.7 %)	86 (27.4 %)
PB3	"I have knowingly submitted someone else's work as my own in the past."	3.33	1.163	8 (2.5 %)	101 (32.2 %)	36 (11.5 %)	117 (37.3 %)	52 (16.6 %)
PB4	"I have been engaged in copying others' works from a digital source (e.g., the Internet) and used them in my own assignments without acknowledgement in the past."	3.48	1.142	7 (2.2 %)	95 (30.3 %)	22 (7.0 %)	127 (40.4 %)	63 (20.1 %)
PB5	"I frequently copied others' works from a digital source (e.g., the Internet) and used them in my assignments with acknowledgement in the past few years."	3.00	1.204	22 (7.0 %)	125 (39.8 %)	34 (10.8 %)	97 (30.9 %)	36 (11.5 %)
PB6	"I have used unauthorized material or resources during exams or assignments in the past."	3.48	1.082	4 (1.3 %)	86 (27.4 %)	25 (8.0 %)	153 (48.7 %)	46 (14.6 %)
Subjective norms								
SN1	"I think most people who are important to me think I should copy others' works from a digital source (e.g., the Internet, ChatGPT) and use them in my own assignments without acknowledgement."	3.19	1.078	7 (2.2 %)	108 (34.4 %)	47 (15.0 %)	123 (39.2 %)	29 (9.2 %)
SN2	"Many students in my program use uncredited digital sources, which tempts me to do the same."	3.53	1.183	14 (4.5 %)	69 (22.0 %)	37 (11.8 %)	125 (39.8 %)	69 (22.0 %)

Table 2: (continued)

	Statements	Mean	Std. deviation	Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree
SN3	"If I use generative AI to copy others' works from digital sources (e.g., the Internet) and submit it in my assignments without acknowledgement, most people who are important to me would approve."	3.48	1.061	10 (3.2 %)	63 (20.1 %)	51 (16.2 %)	146 (46.5 %)	44 (14.0 %)
Information literacy								
IL1	"I am able to appropriately identify the needed information from a question."	3.69	0.941	4 (1.3 %)	39 (12.4 %)	62 (19.7 %)	155 (49.4 %)	54 (17.2 %)
IL2	"I am able to collect/retrieve information in digital environments."	3.76	0.917	6 (1.9 %)	36 (11.5 %)	34 (10.8 %)	189 (60.2 %)	49 (15.6 %)
IL3	"I am able to use ICT appropriately to obtain information."	3.87	0.739	4 (1.3 %)	15 (4.8 %)	39 (12.4 %)	215 (68.5 %)	41 (13.1 %)
IL4	"I am able to interpret, synthesize, summarize, compare, and evaluate information from multiple sources."	3.96	0.602	4 (1.27 %)	11 (3.5 %)	26 (8.28 %)	233 (74.2 %)	40 (12.7 %)
IL5	"I am able to judge the degree to which information is relevant/practical."	3.90	0.791	3 (1.0 %)	18 (5.7 %)	43 (13.7 %)	193 (61.5 %)	57 (18.2 %)
Behavioral intention								
BI1	"I intend to use AI generative tools to create essays, reports, or other coursework without disclosing AI use."	3.47	1.076	20 (6.4 %)	68 (21.7 %)	43 (13.7 %)	129 (41.1 %)	54 (17.2 %)
BI2	"I intend to use AI tools to paraphrase others' work without proper citation to avoid plagiarism detection."	3.48	1.197	11 (3.5 %)	87 (27.7 %)	22 (7.0 %)	129 (41.1 %)	65 (20.7 %)
BI3	"I intend to understand the information and ideas generated by AI chatbots, and present them as my own."	3.44	0.975	22 (7 %)	59 (18.8 %)	42 (13.4 %)	162 (51.6 %)	29 (9.2 %)
BI4	"I intend to use AI to complete assignments when prohibited by course policies."	3.48	1.117	14 (4.5 %)	67 (21.3 %)	33 (10.5 %)	153 (48.7 %)	47 (15.0 %)
Generative AI tools adoption								
AD1	"I regularly use text-generative tools such as ChatGPT to write assignments and presentations without proper disclosure."	3.47	1.105	11 (3.5 %)	74 (23.6 %)	29 (9.2 %)	155 (49.4 %)	45 (14.3 %)
AD2	"I submit AI-generated content as my own original work in most of my courses."	3.21	1.248	11 (3.5 %)	126 (40.1 %)	23 (7.3 %)	95 (30.3 %)	59 (18.8 %)
AD3	"When using AI tools for assignments, I modify the generated content to avoid plagiarism."	3.14	1.090	7 (2.2 %)	120 (38.2 %)	35 (11.1 %)	126 (40.1 %)	26 (8.3 %)
AD4	"I use AI to paraphrase others' work without proper citation to bypass plagiarism detection."	3.23	1.094	6 (1.9 %)	111 (35.4 %)	34 (10.8 %)	132 (42.0 %)	31 (9.9 %)

Scale: from 1 to 5: strongly disagree to strongly agree.

students' intentions behind their use of GenAI (Yang et al. 2021), particularly the factors that may influence their misuse for academic dishonesty, this study shifts its focus beyond previous research, which has primarily concentrated on the adoption of GenAI tools.

Our findings support prior studies that perceived behavioral control (PBC) strongly correlates with moral obligations (MO) (Conner and Armitage 1998). PBC also correlates with past behaviour (PB) (Cronan and Al-Rafee 2008), subjective norms (SN), and information literacy (IL) (Lund et al 2023). Moral obligations (MO) strongly

correlate with past behaviour (PB) (Conner and Armitage 1998; Cronan and Al-Rafee 2008). More obligations (MO) also correlate with subjective norms (SN) and information literacy (IL) (Li et al. 2015; Zwick et al. 2019). Past behaviour (PB) positively correlates with subjective norms (SN) and information literacy (IL) (Danis et al. 2010). Subjective norms (SN) correlate positively with information literacy (IL). The findings show that the TPB constructs (e.g., PBC, SN) show moderate to strong correlations to influence engagement in academic honesty (Uzun and Kilis 2020).

Table 3: Correlations and covariances estimate.

Constructs			Correlation	Estimate	S.E.	C.R.	P
PBC	<-->	MO	0.597	8.898	0.981	9.067	***
PBC	<-->	PB	0.544	9.893	1.170	8.454	***
PBC	<-->	SN	0.618	6.477	0.697	9.296	***
PBC	<-->	IL	0.395	4.191	0.645	6.493	***
MO	<-->	PB	0.609	9.448	1.026	9.204	***
MO	<-->	SN	0.524	4.685	0.571	8.209	***
MO	<-->	IL	0.517	4.684	0.577	8.125	***
PB	<-->	SN	0.621	6.774	0.726	9.332	***
PB	<-->	IL	0.429	4.746	0.680	6.981	***
SN	<-->	IL	0.447	2.852	0.395	7.226	***

Table 4: Validation of the hypotheses.

Hypothesis	Constructs			Estimate	S.E.	C.R.	P-value	Validation
H ₁	BI	<—	PBC	0.223	0.053	4.234	***	Accepted
H ₂	BI	<—	MO	−0.273	0.064	−4.289	***	Accepted
H ₃	BI	<—	PB	0.053	0.051	1.035	0.301	Rejected
H ₄	BI	<—	SN	0.508	0.090	5.644	***	Accepted
H ₅	BI	<—	IL	−0.253	0.075	−3.386	***	Accepted
H ₆	AD	<—	BI	0.651	0.055	11.780	***	Accepted

Standard error (SE); critical ratio (CR). The level of significance is $p < 0.05$, shown as (***), indicating that the p -value is < 0.005 , which is lower than the threshold value of 0.05.

Table 5: Model fit indices values.

Model fit indices	χ^2	DF	GFI	TLI	CFI	NFI	RMSEA
Acceptable fit	$\chi^2 \leq 3$	DF	GFI ≥ 0.90	TLI ≥ 0.90	CFI ≥ 0.90	NFI ≥ 0.90	RMSEA ≤ 0.08
Model fit values	4.075	5	0.982	0.924	0.982	0.977	0.099

Cutt off values citations: Kline (2023); Bentler and Bonett (1980).

Our findings that behavioral intention (BI) is significantly influenced by perceived behavioral control (PBC) (Cronan and Al-Rafee 2008; Stone et al 2010; Camara et al. 2017) and subjective norms (SN) (Mavrinac et al. 2010; Stone et al 2010) are consistent with previous studies. Similarly, our findings validate the findings of previous studies that moral obligation (MO) (Rest et al 1997; Cronan and Al-Rafee 2008) and information literacy (IL) (Celik et al 2021; Zakar et al. 2021; Sriwisathiyakun and Dhamanitayakul 2022) exert a significant negative influence on behavioral intention (BI) to misuse technology for academic dishonesty. Our study's finding that behavioural intention (BI) has a positive significant influence on generative AI tools adoption (AD) to engage in academic dishonesty is aligned with the previous studies (Towler and Shepherd 1991; Godin and Kok 1996; Stone et al 2010; Camara et al. 2017). The goodness of fit indices suggests a favourable fit for the model $\chi^2 = 4.075$,

DF = 5; $p = 0.001$; GFI = 0.982, AGFI = 0.901, NFI, 0.977, RFI, 0.902, IFI = 0.982, TLI = 0.924, and CFI = 0.982, RMSEA = 0.099.

Our findings are comparable with previous studies in that 63 % of respondents think that they have the ability to copy from other sources without giving proper credit (McCabe et al 2002; Mahmud and Bretag 2013; Azam and Naeem 2022). However, the majority (72.3 %) of respondents indicated that it would be difficult for them to use sources in an assignment without citation (Stone et al 2010; He et al. 2024). Our findings indicated that 37 % of nursing students disagree with the notion that academic integrity is essential for personal growth and learning. The apparent contradiction in our findings, where 74 % of respondents value honesty in academic work, yet 54 % admit to submitting someone else's work, reflects a well-documented dissonance between ethical beliefs and behavior in high-pressure academic environments. While our results align with prior

studies (Conner and Armitage 1998; Sulmasy 2000; Dogaš et al. 2016) in affirming nursing students' theoretical commitment to integrity, the gap in practice may stem from contextual factors such as: heavy workload, time constraints, and grade pressure. Moreover, some students may distinguish between “dishonesty” and “collaborative” reuse of others' work, viewing the latter as less severe (Azam and Naeem 2022). Moreover, the majority (63 %) of the respondents have used unauthorized material or resources during exams or completing an assignment in the past. These findings are consistent with the literature (Jeergal et al. 2015; Ip et al. 2016; Carnero et al. 2017; Aikins 2019). Overwhelming respondents (between 58 and 64 %) showed their behavioral intention of copying from other sources without citing these sources in their essays, reports, or other coursework in the near future.

Moreover, our findings indicated that 64 % of nursing students frequently employed AI text-generative tools, such as ChatGPT to write assignments and presentations. Almost 50 % of the nursing students were engaged in malpractices such as cheating on a class test/exam, writing a research paper for another student, and committing plagiarism using AI tools. These findings are comparable with previous studies (Ismail 2018; Błachnio 2019). Our findings revealed a concerning trend of academic dishonesty among nursing students. This may be because students are prioritizing grades over learning which means they might resort to shortcuts like plagiarism or malpractice to achieve good grades without fully understanding the academic topic or a concept. Previously several studies indicated the prevalence of plagiarism in nursing education due to inadequate research skills, time constraints, prevalence of peer cheating, peer pressure and stress, fear of failing, and a lack of awareness regarding citation methods (Cho and Shin 2014; Cho and Hwang 2019; Goodwin and McCarthy 2020; Fadlalmola et al. 2022). Additionally, several studies have indicated that academic dishonesty among students can lead to unethical behavior in clinical settings (Rafati et al. 2020; He et al. 2024).

5.1 Implications of the Study

Our study has strong implications for academicians and librarians to play a vital role in addressing the root causes of plagiarism and potentially unethical behavior, as well as promoting a culture of ethical practices within nursing education. The literature strongly supports the notion that academic staff's commitment to upholding academic honesty plays an important role in reducing a culture of

cheating and academic dishonesty (Dench and Joyce 2022). Moreover, research showed that student perception of faculty commitment to academic integrity influences their own behavior (Godin and Kok 1996). As a result, students are less likely to cheat when they perceive a high risk of detection and consequence; similarly, when they think their teachers take academic dishonesty seriously, they are more likely to adopt fair academic practices themselves. Therefore, there is a need to introduce evidence-based interventions that could influence behavioral intentions of nursing students towards academic integrity.

Our findings revealed that the majority of the nursing students possess a good set of information literacy skills, which is good in terms of recognizing information needs and seeking behavior as well as using information ethically and responsibly. Previous research also supports the effectiveness of information literacy skills in reducing plagiarism (Jiang et al 2013). Librarians have an obligation to equip nursing students with critical evaluation skills to assess sources' credibility (Goodsett and Schmillen 2022), effective research strategies to find and cite reliable information sources efficiently and ethically (Redmond et al. 2020), the use of citation management tools such as Mendeley, Endnote, Zotero, etc. to streamline the citation process and minimize unintentional plagiarism (Rabasa and Abrizah 2022), and plagiarism detecting services, so as to maintain academic integrity in the age of AI generative tools (Liu et al. 2023).

Moreover, there is a need for academicians and librarians to conduct educational initiatives such as workshops or seminars specifically aimed at influencing students' behavior regarding plagiarism (attitudes, subjective norms, perceived behavioral control) and promoting positive attitudes towards ethical practices in nursing education along with introducing strategies for time management and coping with peer pressure, stress, and fear of failure.

Therefore, there is a need that librarians of the participating nursing colleges should develop LibGuides or online research guides specifically tailored to nursing students, containing resources on ethical research, guidelines on citation styles, and plagiarism detective tools and preventive strategies.

5.2 Limitations of the Study

The researchers acknowledge a few limitations of the study, such as the data for this study being collected through a survey. The self-reported data through surveys does have always limitations. In the context of this study, participating students might have responded in a way that they think is

socially acceptable. For example, they might have underestimated instances of cheating or overstating their moral obligations. It is also possible that they forget specific instances of cheating or are unable to recall past behavior, especially if it happened some time ago. They might have interpreted the question statement differently, leading to inconsistencies in responses.

Furthermore, some survey items may have been subject to varying interpretations. While these limitations exist, we tried to minimize them by ensuring respondent anonymity in the questionnaire to encourage honest responses. Moreover, the questionnaire was pilot-tested with a small sample of the target population to refine item clarity and reviewed by two experts for content validity. The questionnaire's reliability was further examined using Cronbach's alpha.

The generalizability of the findings can be noted as a limitation due to differences in contexts, demographic factors, availability of infrastructure, policy guidelines, ethical considerations, and level of AI literacy. Therefore, the findings of the study should be generalized carefully. In order to gain a broader perspective and enhance the generalization, a cross-country study is recommended.

Another limitation may be of measuring and reporting perceived behavioral intentions towards the misuse of GenAI tools for academic dishonesty, rather than attempting to directly observe actual intentional or unintentional misuse. Therefore, the findings primarily reflect participants' perceived intentions, which require further validation through empirical observations and may not perfectly predict actual behavior.

It is also acknowledged that the statements in the survey are not specific to nursing students. In fact, the same survey could be run with other groups (medical students, law students etc.) to generate interesting comparative insights which could also help contextualize the results and draw conclusions.

5.3 Novelty and Originality

This study makes a novel contribution to the literature by investigating the factors that influence students' behavioral intention to adopt generative AI (GenAI) tools for academic dishonesty. While previous research has primarily focused on the adoption of GenAI tools, this study shifts its focus to examine the role of perceived behavioral control, moral obligations, past behavior, and subjective norms in shaping students' intention to misuse GenAI tools for academic dishonesty. Furthermore, it explores the subsequent impact of this intention on the adoption of GenAI tools.

6 Conclusion

This study developed a seven-factor GenAI adoption model grounded in the Theory of Planned Behavior (TPB) to examine its relationship with academic dishonesty among nursing students. The findings indicate that perceived behavioral control (PBC) and subjective norms (SN) strongly predict behavioral intentions, while moral obligations (MO) and information literacy (IL) serve as protective factors for academic integrity. Moreover, behavioral intentions (BI) were found to directly correlate with generative AI tools adoption (AD) for academic dishonesty, highlighting the need for proactive institutional responses.

To bridge the gap between ethical principles and practice, evident in our contradictory findings (e.g., 74 % valuing honesty vs. 54 % admitting to plagiarism), we propose a locally tailored, multi-tiered interventions. These include: (i) AI literacy programs co-developed by academics and librarians, focusing on ethical use, citation management, and critical thinking; (ii) faculty-librarian partnerships to embed integrity training into curricula, promoting librarians' expertise in information ethics; and (iii) stress-reduction initiatives (e.g., time management workshops) to mitigate situational pressures driving misconduct.

To extend this work, we recommend cross-disciplinary validation of the GenAI adoption model (e.g., medicine, engineering) to assess universal versus field-specific predictors. The study also suggests that experimental studies should be carried out to measure the effectiveness of AI literacy interventions in reducing academic dishonesty. There is also a need for policy analysis examining how institutional rules (e.g., grading incentives, GenAI bans) shape misconduct rates.

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